

ESSAY: Discuss our understanding of the brain structures and mechanisms that underlie episodic memories

FEEDBACK:

■■■■■ this is good essay on episodic memory with great reference to the role of the hippocampus – it's clear to me that you've engaged/revised with primary mechanisms such as the trisynaptic pathway, LTP etc. The discussion of dentate gyrus circuitry, mossy fibre plasticity, and CA3 pattern completion is good.

But ! This essay would benefit from a clearer conceptual framing of episodic memory itself. A first-class answer typically begins by situating episodic memory within the broader taxonomy of memory systems (think fractionation of different types of memory between short term and long term) and defining key characteristics to highlight why episodic memory is interesting. The introduction currently focuses almost immediately on hippocampal plasticity, which narrows the scope of your discussion. To strengthen your essay, you could explicitly distinguish episodic memory from other declarative forms (e.g., semantic memory), briefly outline the major brain systems involved (hippocampus, neocortex, prefrontal cortex), and explain that episodic memory requires integration of spatial and temporal information. Including a clearer conceptual definition (e.g. E Tulving's description of episodic memory as the ability to mentally 'relive' past events) would provide a stronger theoretical foundation for the mechanisms discussed later.

A second area for improvement is the integration of mechanisms across levels of explanation. The essay contains detailed descriptions of hippocampal circuitry and molecular plasticity but sometimes treats these elements in isolation rather than showing how they jointly support episodic memory formation. A first class answer would explicitly connect hippocampal computations (pattern separation in dentate gyrus and pattern completion in CA3) with synaptic plasticity mechanisms such as NMDA-dependent LTP and LTD. In addition, the essay would benefit from stronger empirical grounding. Classic experimental evidence (such as Bliss and Lomo's discovery of LTP, NMDA receptor knockout experiments in spatial learning tasks, or lesion studies like H.M) should be used to demonstrate how synaptic plasticity and hippocampal circuits support memory formation. Think about structure-to-function relationships. At the moment, you've got some scientific evidence present, but it's a bit scattered and random; it could be used more strategically to support specific claims that you're making. The discussion of LTP properties (cooperativity, associativity, synaptic specificity) is accurate but could be tied more explicitly to behavioural learning and episodic encoding.

Finally, the essay would reach a higher standard by expanding the systems-level perspective on episodic memory. First-class answers typically emphasise that episodic memory is not stored solely in the hippocampus but emerges from hippocampal-cortical interactions. The current essay briefly mentions cortical consolidation but does not fully develop the concept. Including mechanisms such as hippocampal replay during slow-wave sleep, sharp-wave ripple communication with the cortex, and competing theories of consolidation (e.g., the standard model versus multiple trace theory) would demonstrate deeper understanding. The discussion of pathology (e.g., Alzheimer's disease) is relevant but should be more tightly connected to hippocampal circuitry, (think about where that atrophy first occurs) particularly early degeneration of entorhinal inputs and resulting deficits in pattern separation. In addition, the essay could strengthen its critical evaluation by noting limitations of cell assembly theories or discussing reconstructive aspects of episodic recall and the role of the prefrontal cortex in memory retrieval. Incorporating these broader systems-level processes and critical perspectives would help elevate the essay from a strong mechanistic description to a more comprehensive first-class answer. I hope this helps. I've given some corrections in my comments.

Discuss our understanding of the brain structures and mechanisms that underlie episodic memories

Episodic memories refer to memories of our experiences and life events. They are indexed and temporarily stored by the hippocampus, although they can be consolidated to the cortex over time. Long-term potentiation (LTP) facilitates plasticity in the hippocampus and thus allows for the creation of neural networks, according to the cell assembly model. This results in cell networks being formed in the hippocampus which can be activated in order to retrieve a memory. In cases such as Alzheimer's, LTP and connections between synapses are disrupted which can lead to memory loss. This demonstrates the crucial role of the hippocampus in episodic memory.

The hippocampus is believed to be responsible for the storage and retrieval of episodic memory. Input to the hippocampus is received at the entorhinal cortex, which projects to the dentate gyrus, which connects to the region CA3 via mossy fibres. Entorhinal input diverges onto a large number of granule cells in the dentate gyrus, similar to the divergence of input in the cerebellum, which has led psychologists to theorise that the cellular expansion from the entorhinal cortex to dentate gyrus is responsible for pattern separation and fine discrimination of inputs (Kandel, 2021). This is supported by neurogenesis in the dentate gyrus which enables it to be a highly plastic area specialised for pattern distinction (Christian et al. 2019). The connection between the entorhinal cortex and dentate gyrus is referred to as the perforant pathway and is responsible for ensuring that memories are coded separately. After this pathway, mossy fibres from the dentate gyrus synapse on pyramidal cells at CA3, which project to CA1 via Schaffer collaterals. Overall, this sequence is known as the indirect pathway or trisynaptic pathway. There also exists a direct pathway from the entorhinal cortex to CA3 and CA1. CA2 also provides input to CA1, meaning that CA1 receives three sources of excitatory input. ~~CA2 is believed to play a role in social memory, releasing oxytocin and vasopressin which regulate social behaviour.~~ When CA2 pyramidal neurons were inactivated in an experiment, mice experienced impaired ability to remember conspecifics (Hitti & Siegelbaum, 2014) which demonstrates that CA2 is important for social memory. Each of these structures therefore plays a vital role in memory at the hippocampus.

Commented [EC1]: Yes. Episodic memory is usually defined as memory for personally experienced events that includes contextual information such as T-P-P: Time, Place, and subjective Perspective

Commented [EC2R1]: e.g. consider whether animals have this memory – and to what degree they therefore have autonoetic consciousness

Commented [EC3]: re synaptic plasticity and LTP/LTD: LTP is a candidate synaptic mechanism for learning with the cell assembly concept originating with Hebb. You should separate Hebbian assemblies from modern engram work.

Commented [EC4R3]: Questions: What evidence directly links LTP to memory formation rather than just synaptic strengthening? Are there experiments where blocking LTP impairs memory?

Commented [EC5]: Yes - but episodic memory networks involve hippocampus interacting with distributed cortical ensembles! AT the moment, your sentence implies that memories exist SOLELY in hippocampal networks, which is incorrect. Clarify hippocampus-cortex interactions during retrieval.

Commented [EC6]: well - Alzheimer's pathology involves synaptic loss, tau pathology, amyloid toxicity, inflammation, and network dysfunction across multiple brain regions. Neuropathology in the hippocampus is associated with cell death. LTP disruption is only one mechanism (and is mainly shown in animal models)

Commented [EC7]: A correct conclusion but weak as evidence reasoning. Alzheimer's affects many brain regions beyond the hippocampus - stronger evidence comes from lesion cases (e.g., H.M.), neuroimaging, and animal studies.

Commented [EC8]: Yes, the hippocampus is essential for encoding and retrieval, but long-term storage also resides in distributed cortical networks.

Commented [EC9]: Nice description of the start of the trisynaptic circuit! But do consider that the entorhinal cortex is not simply an input relay; it performs spatial and contextual processing (grid cells, etc.).

Commented [EC10]: good outline of hippocampal circuitry

Commented [EC11]: Yes - this is the temporoammonic pathway, and its functional significance (e.g., comparison of stored vs incoming information) could be mentioned.

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Commented [EC12]: Not relevant in this essay - 'social memory' as a concept is interesting but peripheral & not helpful here unless you explain how CA2 contributes to episodic representation.

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Commented [EC13]: quick summary of specific roles: pattern separation (DG), pattern completion (CA3), comparison/output (CA1).

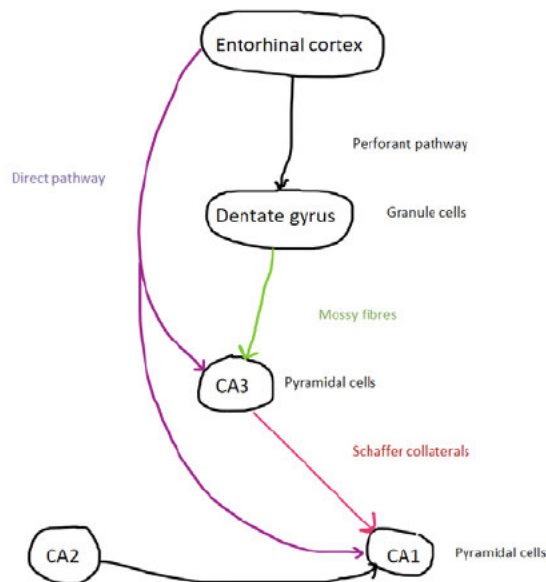


Fig 1: diagram to represent major inputs to CA1 in the hippocampus

On an intracellular level, the hippocampus adapts to experience in order to store episodic memories. Synaptic plasticity at excitatory synapses in the direct and indirect pathways allow for connections between neurons to be strengthened via long-term potentiation (LTP) which causes a synapse to be more likely to fire. LTP is initiated in Schaffer collaterals when the presynaptic and postsynaptic neuron are activated one after another within a short period of time. This is because NMDA receptors at the postsynaptic neuron have an Mg^{2+} block which can only be removed when glutamate binds at the same time as the postsynaptic membrane depolarises, triggering a cascade which causes more AMPA receptors to be produced in a process known as AMPAfication. This ensures that when the presynaptic neuron next releases glutamate, the postsynaptic neuron will be more likely to respond to it. Mossy fibres experience a non-Hebbian type of plasticity by which activation of the postsynaptic cell is not required for the synapse to become stronger, as Ca^{2+} at the presynaptic neuron triggers cAMP and PKA which phosphorylate presynaptic vesicle proteins to enhance the release of glutamate at the synapse.

The existence of LTP is crucial for the formation of episodic memory in the hippocampus due to three features of LTP – cooperativity, associativity, and synapse specificity. Cooperativity refers to the idea that LTP can only occur with near-simultaneous stimulation of many inputs because the Mg^{2+} block requires large

Commented [EC14]: The diagram is useful & beautifully drawn! but the text doesn't explicitly explain what the figure illustrates or why it matters for episodic memory.

Commented [EC15]: Better phrasing would be increased synaptic efficacy rather than this

Commented [EC16]: This describes spike-timing dependent plasticity rather than classical LTP induction protocols.

Commented [EC17]: correct

depolarisation in the postsynaptic neuron to be removed. Therefore, only events which are significant and activate many inputs will cause memory storage (Kandel, 2021). Associativity involves a weak input from a neutral stimulus to be paired with a strong input from an unconditioned stimulus to be paired together as LTP allows for strengthening of synapses between cells involved in both. This enables for associations in the hippocampus between related episodic memories. Associativity may also allow for reward-based learning via classical conditioning. CA1 receives dopaminergic input from the VTA and substantia nigra which can evoke LTP which may cause strengthening of associated synapses (Lisman & Grace, 2005). Finally, synapse specificity – the lack of LTP occurrence at a synapse that is not active – ensures that information that is relevant to an episodic memory will be stored together whereas irrelevant information will not be. Therefore, the LTP is valuable for memory formation in the hippocampus.

Tonegawa et al. (2011) suggested that cells which synapse together in CA1 form a network of neurons which activate together upon memory recall in their Cell Assembly Model. This cell assembly is active again during slow-wave sleep in order to consolidate the episodic memory. They also discovered that dentate gyrus granule neurons in mice which were active during fear encoding could elicit a freeze response when artificially activated again, supporting that cell assemblies exist throughout the hippocampus in order to store episodic memories and that reactivation of cell assemblies can cause memory retrieval. Marr (1971) added to the Cell Assembly Model by suggesting that CA3 is responsible for memory retrieval. This is because a subset of CA3 neurons in a network being activated can cause activation of the entire cell assembly in a pattern completion process due to the recurrent connections which are abundant in CA3. Thus, groups of cells within the hippocampus may form memory engrams.

Disorders in the hippocampus can lead to memory loss and dementia. Alzheimer's, for example, is caused by the accumulation of b-amyloid and tau protein. Hyperphosphorylated tau protein can cause rigidity in the cytoskeleton and formation of 'tau tangles' which form neurofibrillary tangles inside neurons to obstruct synapses. B-amyloid builds up between neurons to disrupt synaptic activity (Abdulkhalik et al., 2026). Tau protein is also implicated with neurodegenerative symptoms of long COVID (Marwaha, 2023). Obstruction of synapses in the hippocampus could cause memory loss because pattern at CA3 would be impaired, for example. A major symptom of Alzheimer's is disorientation which could be associated with damage to synapses in the hippocampus as it is home to place cells which allow for spatial navigation and memory. Lesions to the hippocampus can cause similar effects. For example, in the case of HM (Scoville & Milner, 1957), patient HM was unable to navigate new locations and experienced both anterograde and partial retrograde amnesia (lack of ability to form new memories and ability to retrieve old memories) as his hippocampus was unable to form new memories. However, HM was able to remember some previous

Commented [EC18]: well - The cell assembly model was originally proposed by Hebb (1949). Tonegawa provided experimental evidence for engrams using optogenetics.

Commented [EC19R18]: Hebb's seminal work in synaptic plasticity suggested that tightly connected networks of neurons (ie. Hebbian cell assemblies) are the building blocks of cognitive functions. These assemblies are not necessarily restricted to a small cortical locus but may be dispersed over distant cortical areas.

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Commented [EC21]: Good classical evidence but consider that H.M.'s lesion extended beyond the hippocampus to medial temporal lobe structures (entorhinal cortex, amygdala)

Commented [EC22]: Navigation impairment is not the key feature typically highlighted; the core deficit was forming new declarative memories.

episodic memories which demonstrates that the hippocampus is not solely responsible for memory; more likely, the hippocampus gradually consolidates memories by transferring them to the neocortex, strengthening connections between neurons which were activated when the memory first occurred. Another patient with lesions to the hippocampus, Clive Wearing, experienced both anterograde and retrograde amnesia (Sacks, 2007). This emphasises the importance of the hippocampus for the encoding and retrieval of episodic memory.

In conclusion, episodic memory encoding and recall relies heavily on cell assemblies within the hippocampus which are strengthened by LTP. While memories can be consolidated to the cortex, this process happens over an extended period of time. Various regions of the hippocampus have specific functions in episodic memory: CA2 is involved with social memory, CA3 with pattern completion and memory retrieval and the dentate gyrus with pattern separation. Dementias can reduce the ability of patients to recall episodic memories as b-amyloid and tau protein disfunctions can impair synaptic connections between neurons to weaken connections between cells.

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Suggestions for how to introduce this essay (1st class):

Introduction

Memory (see Fig. 1) can be considered as the encoding, consolidation, storage and retrieval of information. Tulving defines episodic memory as a brain/mind system which allows us to recall past experiences. Episodic memory requires an integration of spatial as well as temporal information. There are several brain structures involved in the processing and retrieval of memory, including the hippocampus, neocortex and prefrontal cortex. One important aspect of episodic memory is autonoetic consciousness which describes a kind of mental time travel where we can re-experience certain past or future events.

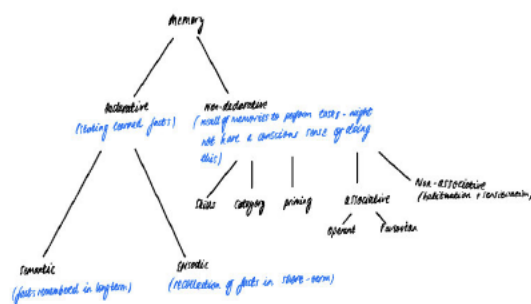
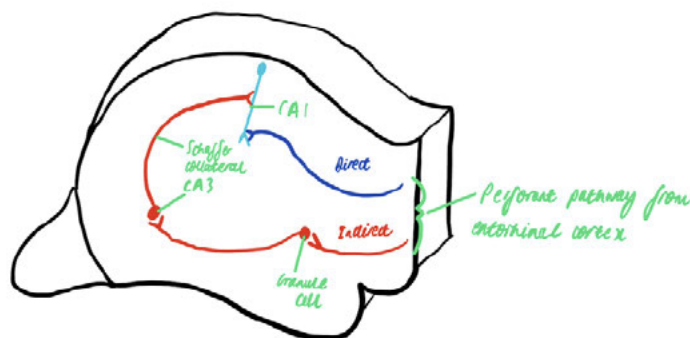


Figure 1 Subdivisions of memory

Role of the hippocampus in episodic memory

The hippocampus and prefrontal cortex are particularly important for declarative (semantic and episodic) memory which is the conscious recall of information. Located deep in the temporal lobe, the hippocampus contains a loop of connections which receive information from the entorhinal cortex. Declarative memory is dependent on a cortico-hippocampal circuit which has two pathways (see Fig. 1). The dentate gyrus facilitates pattern separation of a complex memory in a process known as expansion recoding. By this mechanism, distinct firing patterns of entorhinal input are transformed into activity which is represented across a much larger number of granule cells. Representation of the input in higher-dimensional space means distinct patterns are linearly separable. Interestingly, the dentate gyrus is one of the few sites of adult neurogenesis. In fact, formation of new granule cells has been shown to increase the ability of pattern separation. Evidence for the role of the hippocampus in episodic memory comes from lesion studies such as the case of H.M. H.M. had debilitating epilepsy and underwent bilateral removal of the hippocampus which resulted in a graded retrograde amnesia, where the most recent memories were lost, total anterograde amnesia but a retained procedural memory. This suggests that the hippocampus plays a role in processing declarative memories and that the hippocampus is not required for all types of memory (e.g. procedural).



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